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Session: Disruptive Innovations for Utilities

Digital Twins for Distribution Utilities: From Grid Visibility to Operational Foresight

Presented By

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Transformation in the Distribution Networks

- Rising DER and EV penetration, increasing bi-directional power flows
- Investing heavily in digital infrastructure such as AMI systems, GIS platforms and customer management systems
- Traditional monitoring tools & Static dashboards are no longer sufficient
- Need for real-time operational insights

CONTEXT - NEED FOR A UTILITY-CENTRIC DIGITAL TWIN

Why Existing Systems are not Sufficient?

- AMI, GIS, and CRM systems operate independently – *fragmented view of the grid*
- SCADA visibility is largely limited to substations and major feeders
- Cross-system analysis requires manual reconciliation
- Limited capability for predictive or scenario-driven operations

- Most digital twin research has focused on the transmission systems or asset-level modelling

RELEVANCE – UTILITY-CENTRIC DIGITAL TWIN

Cost-effective, immediately deployable system

- Digital twin framework tailored for low and medium-voltage grids
- Leverages existing systems such as MDMS, HES, CIS, GIS
- No emphasis on additional field instrumentation or custom sensors

- Offers a real-time, data-driven approach to monitor, simulate and manage grid operations with high accuracy

DIGITAL TWIN ARCHITECTURE - *Proposed Architecture*

A 5-layer architecture built for utility scale and speed

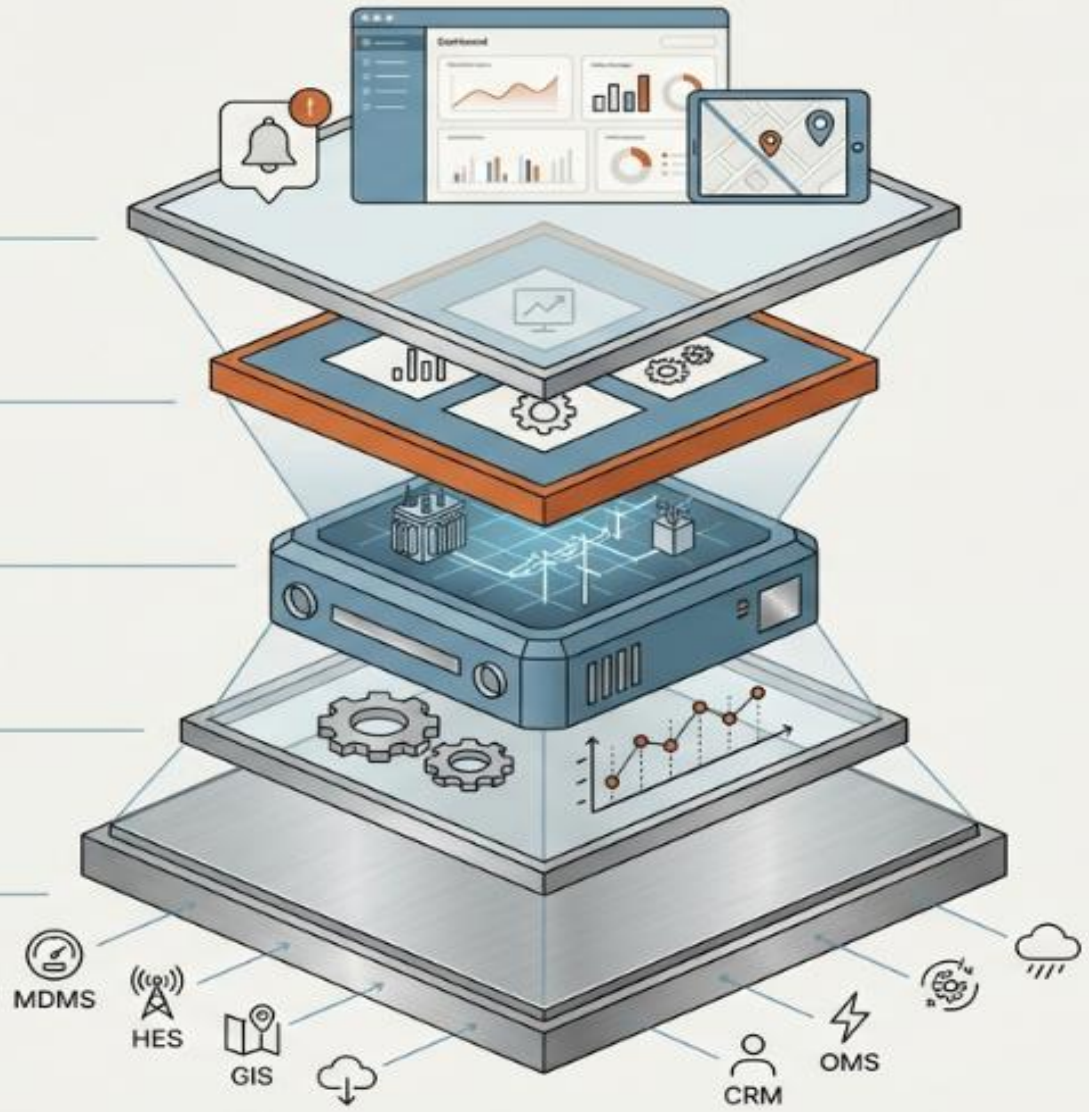
5. Visualisation & Insights
Actionable dashboards and alerts for utility managers and field operators.

4. Use Case Layer
The operational hub supporting live analytics, simulations, and planning.

3. Core Digital Twin Engine
Real-time, living models of transformers, feeders, and the LT network.

2. Data Synchronisation
Precise temporal alignment of time-series data with grid topology.

1. Data Ingestion
Raw inputs from MDMS, HES, GIS, CRM, OMS, and weather systems.



KEY OPERATIONAL USE CASES

Operational Applications

- Load Forecasting & Peak Management
  - *Analyses real-time meter data to forecast transformer or feeder level demand*
- Rapid fault localisation and outage impact estimation
  - *Integrating GIS & AMI data, identifies likely fault locations and affected zones*
- Transformer Health & Overload Detection
  - *Use aggregated meter data to monitor transformer stress and predict overloading*
- DER hosting capacity simulation
  - *Model solar and EV impacts under various scenarios to assess grid stability*
- Theft Detection & Loss Analytics
  - *Mismatch analysis between technical and billing data highlights potential theft or configuration errors*
- Infrastructure planning through network simulation
  - *Utilities can simulate changes like adding feeders, shifting loads, or responding to EV or battery storage integration*

CONSIDERATIONS & CHALLENGES

- Data Quality & Availability
  - *Struggle with poor meter communication, outdated asset records, duplicate master data, etc.*
- Multi-vendor System Integration
  - *Integrating diverse vendor systems is complex*
- Real-Time vs Near Real-Time Processing
  - *Use cases vary in latency needs. Ex: fault detection (real-time), forecasting (scheduled processing)*
- User Readiness
  - *Adoption needs training, change management and role-based workflows*
- Scalability
  - *Rural and smaller utilities may require lighter configurations*
- Cybersecurity
  - *With IT-OT integration, security is vital. RBAC, encryption, and CERT-IN compliance*

CONCLUSION

Enabling Operational Digital Twins

- Integrates existing utility systems into a unified operational model
- Enhances LV/MV grid visibility
- Supports predictive maintenance and outage preparedness
- Improves DER and infrastructure planning
- Deployable using current infrastructure
- Foundation for the AI-driven complete grid intelligence

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THANK YOU

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Links/References (If any)